

Least-Norm Source Budgets and Native-Ray Obstruction in Literal Twin-Prime Shells

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Abstract

TPC-295 proved that the unrestricted finite source-correlation map attached to a physical prime shell is onto, but existence alone does not measure its cost. This paper derives the exact least-norm source budget for a target and an exact source-cost/physical-energy tradeoff. If A contains the physical shell vectors and $G = A^\top A$, then the minimum source norm for $A^\top h = b$ is $b^\top G^{-1}b$, attained by $h = AG^{-1}b$. On the inherited 18-row literal grid, a 70-digit replay finds that all 18 TPC-294 weighted minimizers have cost below $10^{-3} \|\beta\|^2$ in the unrestricted source space. In contrast, even the enlarged one-dimensional proxy generated by the frozen source vector has normalized least-squares residual at least 0.9 on all 18 weighted targets. The finite evidence therefore moves the obstruction from raw unrestricted source energy to the dimension and geometry of the native profile. No growing, arithmetic L^2 , or twin-prime conclusion follows.

1 Position on the route

This release remains in the same literal prime-shell dynamical family. TPC-293 compiled pairwise sign preferences into a whole-shell signed graph, and TPC-294 restored the physical Gram magnitudes and found a global weighted sign optimizer on each inherited row [2, 1]. TPC-295 then proved that these ambient targets have finite rational source preimages whenever the shell Gram matrix is nonsingular [3].

The remaining question is quantitative: is the preimage affordable, and is it aligned with the source family that generated the physical vectors? The first part has an exact answer. The second is tested here only through a declared one-dimensional proxy, so that a finite obstruction is not mistaken for a theorem about all native profiles.

2 Physical shell and source correlation

Let $I = I_N$ be the frozen integer interval. For an odd prime q , height H , and exponent s , write

$$K_{H,s}(v) = \frac{H^{2s}}{(H^2 + v^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The deleted-diagonal physical component inherited from the frozen source vector β is

$$(g_q)_u = \sum_{t \in I, t \neq u} q K_{H,s}(u-t) B_q(u, t) \beta_t, \quad u \in I.$$

For $S = \{q : Q < q \leq 2Q\}$, put

$$A = [g_q]_{q \in S} \in \mathbb{Q}^{I \times S}, \quad G = A^\top A, \quad \text{quad} C = A^\top : \mathbb{Q}^I \rightarrow \mathbb{Q}^S.$$

For a target $b \in \mathbb{Q}^S$, its unrestricted source cost is

$$S_A(b) = \min\{\|h\|_2^2 : A^\top h = b\}.$$

The finite source domain in this paper is the full rational coordinate space; the original arithmetic profile is not silently identified with it.

3 Exact source-budget theorem

Theorem 1 (Least-norm source budget). *Suppose $A \in \mathbb{R}^{n \times m}$ has full column rank and $G = A^\top A$. For every $b \in \mathbb{R}^m$, the constraint $A^\top h = b$ is feasible and*

$$S_A(b) = b^\top G^{-1} b, \quad h_b = AG^{-1} b$$

is its unique minimum-norm solution. Consequently, for any $B \geq 0$, there is a solution with $\|h\|_2^2 \leq B$ if and only if $b^\top G^{-1} b \leq B$.

Proof. The vector $h_b = AG^{-1} b$ satisfies $A^\top h_b = GG^{-1} b = b$. Every other solution is $h_b + v$ with $v \in \ker(A^\top)$. Since $h_b \in \text{col}(A)$ and $\text{col}(A) = \ker(A^\top)^\perp$, Pythagoras gives

$$\|h_b + v\|_2^2 = \|h_b\|_2^2 + \|v\|_2^2.$$

Finally, $\|h_b\|_2^2 = b^\top G^{-1} A^\top AG^{-1} b = b^\top G^{-1} b$. The budget statement follows from minimality. $\square$

Proposition 1 (Source-cost/physical-energy tradeoff). *Under the hypotheses of the theorem, define*

$$E_A(b) = \|Ab\|_2^2 = b^\top G b.$$

Then

$$S_A(b)E_A(b) \geq (b^\top b)^2.$$

For a sign target in $\{-1, +1\}^m$, the right-hand side is m^2 .

Proof. Apply Cauchy–Schwarz to $G^{1/2}b$ and $G^{-1/2}b$. Their inner product is $b^\top b$, while their squared norms are $E_A(b)$ and $S_A(b)$. $\square$

The inequality is a useful interface statement: physical cancellation and source affordability cannot both be made arbitrarily strong for the same target. It is exact linear algebra, not an arithmetic estimate.

4 A declared native-ray diagnostic

Let β be the frozen source vector and define its correlation vector

$$v = A^\top \beta.$$

We enlarge the literal frozen direction to the ray proxy $\{\alpha v : \alpha \in \mathbb{R}\}$. This is intentionally generous: failure of the proxy does not say that a richer source family fails.

Proposition 2 (Least-squares distance to the source ray). *If $v \neq 0$, then for any target b*

$$\min_{\alpha \in \mathbb{R}} \|\alpha v - b\|_2^2 = \|b\|_2^2 - \frac{(v^\top b)^2}{v^\top v}.$$

For a sign target, the square root after division by $\sqrt{m}$ is its normalized ray RMS residual.

Proof. Complete the square in $\alpha^2 v^\top v - 2\alpha v^\top b + b^\top b$. $\square$

5 Finite protocol

The audit inherits eight $s = 2$ growth rows, four $s = 1$ exponent-crossover rows, and six $s = 2$ source-control rows from TPC-295. On each row, the physical columns are first accumulated exactly as rational numbers. The producer then converts them to 70-digit arithmetic, diagonalizes G , and solves for the TPC-294 weighted minimizer, the unit-edge max-cut target, and the all-positive target. It records conservative decimal enclosures for cost, energy, condition number, trade product, and ray residual.

The independent replay changes the accumulation order: it builds each column source-first and does not import the producer. It checks the source correlation residual and the identity $\|AG^{-1}b\|_2^2 = b^\top G^{-1}b$. Exact rational fixtures attack both full-rank and duplicated-column singular cases.

The declared diagnostics are

$$\tau(b) = \frac{S_A(b)}{\|\beta\|_2^2}, \text{quad}\tau(b) < 10^{-3}, \text{quadr}_{\text{ray}}(b) \geq 0.9.$$

They are finite modeling choices and carry no exponent credit.

Table 1: Finite source-budget and one-ray audit.

quantity	count
rows	18
shell edges	1,380
weighted-minimum $\tau < 10^{-3}$	18 / 18
max-cut $\tau < 10^{-3}$	18 / 18
all-positive $\tau < 10^{-3}$	18 / 18
weighted-minimum ray RMS ≥ 0.9	18 / 18
max-cut ray RMS ≥ 0.9	18 / 18
all-positive ray RMS ≤ 0.4	18 / 18
tradeoff failures	0 / 54

6 Cost and profile atlas

The headline counts are given in Table 1.

The largest weighted-minimum budget ratio is 0.000584056544646..., and the smallest weighted-minimum ray RMS is 0.912134981827.... Thus the unrestricted source budget diagnostic is passed with room on every row, while the frozen-beta ray diagnostic fails uniformly for the weighted targets. The largest observed Gram condition number is 2497.29180077....

Table 2: Representative high-precision enclosures, displayed by midpoint.

axis	N	H	Q	s	$ S $	$\tau_{\min}$	$r_{\min}$	cond
growth	128	24	9	2	3	0.0001219	0.9121	5.57
growth	192	32	16	2	5	0.0000719	0.9860	14.06
growth	384	50	40	2	10	0.0000462	0.99999	86.71
growth	512	58	60	2	13	0.0000827	0.9958	405.27
growth	512	58	90	2	17	0.0002702	0.9966	2497.29
crossover	256	38	27	1	7	0.0000425	0.9885	4.69
control	384	48	70	2	15	0.0005841	0.9967	1574.62
control	384	52	70	2	15	0.0005577	0.9974	1387.40

At the late 17-prime row, the weighted target has least source cost 0.0225929240206... and cost ratio 0.0002701744808..., while its one-ray RMS residual is 0.9965754488.... At the source-control row with $H = 48$, the cost ratio reaches the finite maximum 0.0005840565446..., but the ray RMS is still 0.9966625161.... The exact tradeoff product over all three target families and all rows lies between approximately 1.04038 and 4.17371, consistent with the exact lower bound one.

7 Interpretation and route boundary

TPC-295's broad image question is therefore not obstructed by a large unrestricted least-norm witness on this finite grid. This is a useful positive result: one cannot reject the weighted sign direction merely on the basis of ambient source energy. The same data reveal a different obstruction. The weighted minimizer is nearly orthogonal, in correlation space, to the frozen-beta ray on every row. A one-dimensional native proxy is consequently too small or points in the wrong direction for these targets.

The conclusion is deliberately limited. The actual Mobius/comparison source class may have several independent directions, and the present calculation does not determine its dimension or image. The source-control cutoff triples also coincide in this physical construction because the frozen β used for the columns is unchanged; this is a modeling boundary, not cutoff-uniform arithmetic evidence. Finally, a finite condition-number range is not a growing-shell bound.

The claim ceiling is thus

$$\text{PROVED_EXACT_FINITE} + \text{NUMERICALLY_CERTIFIED_FINITE} + \text{NUMERICAL_OBSERVATION},$$

with fixed-power credit zero. Literal arithmetic L^2 , a restricted profile theorem, Gate B, and the twin-prime endpoint remain open.

8 Reproducibility and conclusion

From the project directory, run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc296_source_norm_budget_certificate.py --check
python -B experiments/tpc296_independent_checker.py
python -B experiments/tpc296_budget_stress.py
```

The optimized versions use Python's `-O` flag and must have empty standard error and byte-identical standard output. The canonical record is `results/tpc296_certificate.json`.

The Session-named Route-A/Route-B evaluator files are absent from this checkout. The proof package, canonical certificate, independent replay, exact stress suite, PDF audit, and local Bridge-B checker are the available fail-closed evidence; no official evaluator pass is claimed.

The next route question is a restricted-profile basis and growing budget test, not another ambient sign enumeration. This is the `ROUND2_CLUE` recorded by the certificate.

References

- [1] Liang Wang. Magnitude-weighted signed rayleigh cancellation in finite prime shells, 2026. TPC-294 project release in the same research repository.
- [2] Liang Wang. A signed max-cut obstruction for multi-prime cancellation, 2026. TPC-293 project release in the same research repository.
- [3] Liang Wang. Source-correlation image and finite signed feasibility in literal twin-prime shells, 2026. TPC-295 project release in the same research repository.